

Operator:

In []: An operator **is** a symbol that performs an operation on one **or** more operands(val

Types of Operator:- ¶

In []:

1. Arithmetic Opearator
2. Assignment Operator
3. Comaparision Operator
4. Logical Operator
5. Membership Operator
6. IdentitynOperator
7. Bitwise Operator

1.Arithmetic Opearator

In []:

- + >> Addition
- >>Substraction
- * >> Multiplication
- / >>Division
- //>>Floor Division
- % >> Modulus
- ** >>Exponent

1. Addition

In [1]:

```
a=100
b=50
print(a+b)
```

150

2. subtraction:

In [3]:

```
Money=1000
spent=500
print(Money-spent)
```

500

3. Multiplication

```
In [4]: pens=10  
price=15  
print(pens*price)
```

150

4. Division

```
In [5]: print(8/4)
```

2.0

```
In [6]: print(9/2)
```

4.5

5.Floor Division

```
In [7]: print(10//3)
```

3

6.Modulus:

```
In [8]: print(10%3)
```

1

```
In [9]: num=12  
print(num%2)
```

0

7.Exponent:

```
In [10]: print(2**3)
```

8

```
In [11]: print(3**4)
```

81

```
In [12]: print(2**3)
```

8

Arithmetic Mini Project

```
In [14]: num1=int(input("Enter First number:-"))
num2=int(input("Enter Second number:-"))

print("Addition:",num1+num2)
print("Substraction:",num1-num2)
print("Multiplication:",num1*num2)
print("Division:",num1/num2)
print("Floor Division:",num1//num2)
print("Mudulus:",num1%num2)
print("Exponent:",num1**num2)
```

```
Enter First number:- 10
Enter Second number:- 2
```

```
Addition: 12
Substraction: 8
Multiplication: 20
Division: 5.0
Floor Division: 5
Mudulus: 0
Exponent: 100
```

2. Assignment Operator

```
In [ ]: =
+=
-=
*=
/=
%=
```

```
In [16]: s1="Python"
s1*5
```

```
Out[16]: 'PythonPythonPythonPythonPython'
```

```
In [17]: s1="python"
s2="class"
s1+' '+s2
```

```
Out[17]: 'python class'
```

+=

```
In [18]: x=5
x+=3
x
```

```
Out[18]: 8
```

Substract and Assign

```
In [19]: x=5  
x-=2 #x=x-2  
x
```

Out[19]: 3

Multiply and Assign

```
In [20]: X=5  
X*=2 #X=X*2  
X
```

Out[20]: 10

Divide and Assign

```
In [21]: x=5  
x/=4  
x
```

Out[21]: 1.25

Mudulus And Assign

```
In [27]: x=5  
x%=3  
x
```

Out[27]: 2

3. Comaparision Operator

```
In [ ]: == >> Equal  
!= >> Not equal  
> >> Greater than  
< >> less than  
>= >> Greater than equal  
<= >> Less than equal
```

```
In [28]: a=10
b=20

print(a==b)
print(a!=b)
print(a>b)
print(a<b)
```

False
True
False
True

Voting eligibility

```
In [29]: age=20
print(age>=18)
```

True

```
In [30]: age=17
print(age>=18)
```

False

Even and Odd numbers

```
In [31]: x=45
if x%2==0:
    print(f"{x} is an even number")
else:
    print(f"{x} is an odd number")
```

45 is an odd number

```
In [32]: list1=[4,5,20,3,5,6,67,45]
for i in list1:
    if i%2==0:
        print(i,":even")
    else:
        print(i,":odd")
```

4 :even
5 :odd
20 :even
3 :odd
5 :odd
6 :even
67 :odd
45 :odd

```
In [33]: x=34
if x%2!=1:
    print(f"{x} is an even number")
else:
    print(f"{x} is an odd number")
```

34 is an even number

```
In [35]: count=0
for i in 'python':
    print(count,i)
    count=count+1
```

0 p
1 y
2 t
3 h
4 o
5 n

Membership operator

```
In [ ]: in>> Evaluates to True if the specified value is found inside the collection
not in>>Evaluates to True if the specified value is missing from the collection
```

```
In [36]: String ="Python and Data Science"
"Python" in String
```

Out[36]: True

```
In [39]: String ="Python and Data Science"
"Machine Learning" not in String
```

Out[39]: True

```
In [40]: string="Machine learning and Data science"
"Machine learning" in string
```

Out[40]: True

```
In [41]: list1=[4,5,6,10,20,30]
x=5
if x in list1:
    print(f"{x} is present")
else:
    print(f"{x} is not present")
```

5 is present

Identity Operator

```
In [ ]: is  
is not
```

```
In [42]: a=10  
b=11  
print(id(a))  
print(id(b))  
  
140726077303512  
140726077303544
```

```
In [43]: a=10  
b=10  
id(a)==id(b)
```

Out[43]: True

```
In [44]: lista=[1,2,3]  
listb=[1,2,3]  
listc=lista #c points to the same object as a  
  
print(lista is listc)  
print(lista is listb)  
print(lista is not listb)  
  
True  
False  
True
```

Bitwise Operator

```
In [ ]: & :Bitwise And  
| : Bitiwise Or  
^ :Bitwise XOR  
>> :Right Shift  
<< :Left shift
```

```
In [45]: a=10  
bin(a)
```

Out[45]: '0b1010'

```
In [46]: a=29  
bin(a)
```

Out[46]: '0b11101'

```
In [48]: a=11  
        bin(a)
```

```
Out[48]: '0b1011'
```

```
In [49]: b=13  
        bin(b)
```

```
Out[49]: '0b1101'
```

```
In [50]: a=11  
        b=13  
        print(a,bin(a).replace('0b',''))  
        print(b,bin(b).replace('0b',''))
```

```
11 1011  
13 1101
```

Bitwise AND

```
In [ ]: 1 & 1 >> 1  
        1 & 0 >> 0  
        0 & 1 >> 0  
        0 & 0 >> 0
```

```
In [51]: 11&13
```

```
Out[51]: 9
```

Bitwise OR

```
In [ ]: 1 | 1 >> 1  
        1 | 0 >> 1  
        0 | 1 >> 1  
        0 | 0 >> 0
```

```
In [ ]: 1011>>11  
        1101>>13  
  
        1111>15
```

```
In [52]: 11|13
```

```
Out[52]: 15
```


Bitwise XOR

```
In [ ]: 1^1>> 0
        1^0>>1
        0^1>>1
        0^0>>0
```

```
In [ ]: 1011>>11
        1101>>13

        0110>>6
```

```
In [53]: 11^13
```

```
Out[53]: 6
```

Right Shift



```
In [ ]: 11000>>24
```

```
In [55]: 24>>1
```

```
Out[55]: 12
```

```
In [ ]: 1101>>13
```

```
In [56]: 13>>1
```

```
Out[56]: 6
```

Left shift

```
In [58]: 13<<1
```

```
Out[58]: 26
```

```
In [ ]: Task1: Shopping Bill calculator
        Task 2 : student Result
```

